Response to reviewers for BIBE paper 54

Cross Gesture Transfer After Single Gesture Calibration in High Density Electromyography
Charles Lu

Thank you for the detailed reviews. The revised paper adds an error decomposition and calibration-specific confusion matrices, participant sign-flip analyses, an identity-regularized safeguard, and two preprocessing sensitivity checks. It also narrows the claims and clarifies recording cost. Added analyses are explicitly post-hoc or exploratory. Original numerical results are preserved. Yellow highlighting in the marked manuscript identifies new or changed text and table entries; changed figures are identified by highlighted captions. A clean copy and supplementary material are also provided.

Reviewer 1 major comment 1 Error structure

Section IV.D and Table V partition other-gesture trials into correct predictions, errors to the calibration label, and errors to other incorrect labels. Spatial + gain assigns 36.46% of these trials to the calibration label, compared with 5.61% frozen. Supplementary Figs. S2–S7 provide matrices separately by calibration gesture for all six methods, with participant-preserving aggregation. We distinguish output bias from an unproven representation-level mechanism.

Reviewer 1 major comment 2 Single class objective degeneracy

Sections III.C and V now state the cross-entropy objective and its seven affine parameters. We explain why observing only one label does not penalize incorrect decisions on omitted classes, while distinguishing unrestricted classifier collapse from bounded spatial adaptation. Saved logs show zero computed minimum loss in 188/640 spatial and 292/640 spatial + gain cases. We discuss finite-precision saturation and connect the result to the conditional gain/spatial identifiability example without treating exact ties alone as proof.

Reviewer 1 major comment 3 A simple safeguard

We added a fixed identity penalty to both spatial candidate selection and optimization. The penalty averages squared physical-parameter deviations normalized by their allowed ranges; its weight is 1 without an outcome-based sweep. On the original fixed pair 1→2, spatial other-class accuracy rises from 50.50% to 56.30%, and spatial + gain from 36.11% to 37.95%. The latter still falls below frozen at 45.49%. Sections III.E, IV.E, and V and Supplementary Sections S1/S3/S4 explain the improvement, calibrated-class tradeoff, and limited scope. This is an exploratory follow-up on reused participants, not a validated remedy or all-pair result.

Reviewer 1 major comment 4 Generalisability and statistics

The abstract, introduction, results, and conclusion explicitly restrict the findings to four participants, one CSL-HDEMG task, and one implementation. We retain per-participant effects and add exact two-sided sign-flip tests across all 16 sign assignments. Spatial-related contrasts have $p = 0.125$; gain versus frozen has $p = 0.625$. We explain the assumptions and coarse resolution and label bootstrap intervals descriptive. We did not add a second multi-session HD-EMG dataset; this remains a limitation requiring independent replication. We do not substitute the later, separate GRABMyo/SeNic study for that replication.

Reviewer 1 major comment 5 Preprocessing sensitivity

Two revision runs isolate activation of mean-square baseline subtraction under provider mapping and use of the literal upstream mapping under inactive subtraction. Each retrains the source model and uses a matched frozen reference. Spatial + gain reaches 36.11% versus 45.14% frozen with subtraction active, and 37.45% versus 43.75% frozen with literal mapping. The combined other-class deficit persists on the tested pair, while absolute values change. Sections III.E/IV.E and Supplementary S1/S3/S4 disclose that these 32-case checks cover one pair, not all 80 participant-pairs; the literal mapping includes upstream time reversal and is not a full reproduction of every upstream setting.

Reviewer 1 minor comment 1 Density and SAL description

Long verification counts and hashes move to Supplementary S4 and the analysis package. The main discussion is condensed, the seven SAL parameters and objective are explained in Section III.C, and Supplementary Fig. S1 shows the pipeline, grid, and provider gesture IDs. We use verified IDs rather than inventing gesture-name mappings.

Reviewer 1 minor comment 2 Analysis release
A separate BIBE-specific analysis package is included with this revision. Its three reproduction scripts pass and the exact numerical inputs are present. The existing public repository describes the later manuscript and does not contain the original BIBE result bundle, so it is not cited as if it reproduced this paper. Public publication of the separate package is awaiting author authorization; this draft does not claim that publication has occurred. This release request remains open until the author approves publication or supplies an explicit reason for keeping the package nonpublic.

Reviewer 1 minor comment 3 Early definitions
The opening introduction defines calibration-frame accuracy and distinguishes it from held-out calibrated-class accuracy. Negative transfer is defined there as lower overall trial accuracy than the matched frozen recognizer; other-class degradation is named separately.

Reviewer 1 minor comment 4 Table IV frozen reference
The Table IV note now explicitly contrasts 45.49% for session pair 1→2 with the 60.71% all-pair frozen mean and states the restricted 32-case scope.

Reviewer 2 major comment 1 Four participant evaluation
The abstract, introduction, results, limitations, and conclusion now keep the cohort and implementation scope explicit. No universal algorithmic failure, clinical effect, or population significance is claimed. The new sensitivity checks reuse the four participants and are labeled accordingly.

Reviewer 2 major comment 2 Implementation dependence
The abstract and introduction state that only one fixed implementation is evaluated, and Section IV attaches results to that scope. The limitations explain that architecture, bounds, objective, and optimizer changes could mitigate or eliminate degradation. Follow-up preprocessing checks provide limited sensitivity evidence without claiming all-pair or cross-implementation robustness.

Reviewer 2 major comment 3 Public analysis scripts
A separate BIBE-specific analysis package is included with this revision. Its three reproduction scripts pass and the exact numerical inputs are present. The existing public repository describes the later manuscript and does not contain the original BIBE result bundle, so it is not cited as if it reproduced this paper. Public publication of the separate package is awaiting author authorization; this draft does not claim that publication has occurred. This release request remains open until the author approves publication or supplies an explicit reason for keeping the package nonpublic.

Reviewer 2 major comment 4 Recording burden and offline use
Sections III.B and V make the 3 seconds of labeled activity plus 90 seconds of target rest explicit. The discussion notes the thirtyfold rest-to-labeled duration, the implications for rapid recalibration, and the absence of causal streaming, closed-loop feedback, fatigue, and user-adaptation evaluation.

Reviewer 2 minor comment 1 Discussion readability
The discussion is reorganized into four focused paragraphs covering error attraction, single-class loss/identifiability, the safeguard tradeoff, and practical recording/online limits. Detailed verification is moved out of the main text.

Reviewer 2 minor comment 2 Figure 2 readability
Figure 2 uses larger labels, shorter legend text, and a thicker black zero line. The near-zero bar now retains its actual length with a small location marker; the caption explains the encoding.

Reviewer 2 minor comment 3 Descriptive bootstrap intervals
Section III.D explains the four-participant bootstrap limitation and unequal probabilities of the 35 count compositions. Section IV.A repeats that the intervals are descriptive and are not precise population uncertainty bounds. Participant sign consistency and coarse exact-test results are reported alongside them.